

User Interaction

- Most programs rely on user interaction to:
 - Input data.
 - Make decisions or choices.
 - Confirm options for program execution.
- User interaction influences:
 - Data processing.
 - Conditional actions.

Examples

- **Calculator Program:**
 - User inputs numbers and decides operations (e.g., sum, subtract).
- **JavaScript Programs:**
 - **Client-Side (Browser):**
 - Requires constant user interaction (e.g., forms, buttons).
 - **Server-Side (Node.js):**
 - Typically lacks user interaction during execution.
 - Uses configuration files or command-line arguments for inputs.

Client-Side Interaction

- HTML components like:
 - `<input>` for user input fields.
 - `<button>` for actions.
- Interaction involves JavaScript accessing and manipulating the DOM.

Example:

html

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```
<!DOCTYPE html>
```

```
<html>
```

```
<body>

  <input id="myId" type="text"></input>

  <button onclick="console.log(document.getElementById('myId').value)">Get Text</button>

</body>

</html>
```

Dialog Boxes

Dialog boxes are browser-integrated popup windows that interrupt interaction with the webpage. Types include:

1. Alert Dialog Box:

- Displays a message with an **OK** button.
- Code execution halts until closed.
- Syntax:

javascript

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```
alert("Hello, World!");
```

2. Confirm Dialog Box:

- Displays a message with **OK** and **Cancel** buttons.
- Returns a Boolean value:
 - true for **OK**.
 - false for **Cancel**.
- Example:

javascript

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```
let decision = confirm("Are you sure?");
console.log(decision);
```

3. Prompt Dialog Box:

- Displays a message, a text field, and **OK/Cancel** buttons.
- Returns user input as a string or null if canceled.

- Syntax:

javascript

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```
let name = prompt("What is your name?", "John Doe");
```

Summary

- Dialog boxes are basic yet effective for learning and limited use cases.
 - They allow programs to retrieve data and incorporate user decisions.
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Quiz: Interaction with the User

1. **What is the purpose of user interaction in programs?**

- a) To crash the program.
- b) To influence data and execution.
- c) To execute without inputs.
- d) None of the above.

Answer: b) To influence data and execution.

2. **Which type of JavaScript programs often lack user interaction?**

- a) Client-side programs.
- b) Server-side programs.
- c) Both.
- d) None.

Answer: b) Server-side programs.

3. **What does the alert dialog box do?**

- a) Displays a message and halts execution until closed.
- b) Collects user input.
- c) Executes commands silently.
- d) None of the above.

Answer: a) Displays a message and halts execution until closed.

4. **Which dialog box allows user input via a text field?**

- a) Alert.
- b) Confirm.
- c) Prompt.
- d) All of the above.

Answer: c) Prompt.

5. **What will `confirm("Proceed?")` return if the user clicks "Cancel"?**
- a) true
 - b) false
 - c) null
 - d) undefined

Answer: b) false.